

INFINITY FITNESS STUDIOS

Data Model & DAX Measures

Member Retention, Utilization & Operational Analytics
December 2025 - February 2026

Document Control	Details
Document ID	IFS-DM-DAX-001
Version	1.0
Role	Data Analyst
Purpose	Document the portfolio data-model design and DAX measure specifications supporting the confirmed Power BI analysis.
Source Basis	Confirmed Excel-based member, membership, attendance, payment, trainer, class/session, revenue and expense data domains.
Design Status	Logical portfolio reconstruction. Exact historical Power BI table names, relationship directions and DAX code are not retained.
Control	Measures below are defensible calculation specifications for rebuilding the portfolio solution; they are not represented as verbatim historical production DAX.

1. Data Model Objective

The Power BI model should support reliable analysis of retention, attrition, cohorts, renewals, attendance/utilization, membership plans, trainer activity and commercial performance while minimizing double-counting and ambiguous relationships.

2. Recommended Logical Model

Table	Role	Grain / Intended Record	Primary Use
DimMember	Dimension	One logical row per member	Member identity and segmentation
DimPlan	Dimension	One row per membership-plan category	Plan comparisons and filtering
DimTrainer	Dimension	One row per trainer	Trainer filtering and utilization
DimDate	Dimension	One row per calendar date	Time intelligence and period filtering
FactMembership	Fact / lifecycle	One row per membership period/record where source supports it	Joining, renewal, tenure and plan analysis
FactAttendance	Fact	One row per valid check-in / attendance record	Engagement and utilization
FactPayment	Fact	One row per valid payment record	Payment and revenue-related analysis
FactSession	Fact	One row per session / class booking record where available	Trainer activity, bookings and attendance
FactRevenue	Fact	One row per revenue record or period classification	Commercial analysis
FactExpense	Fact	One row per expense record or period classification	Expense analysis

3. Relationship Design

From	To	Relationship	Design Intent
DimMember	FactMembership	1 : many	A member may have multiple membership periods/records.
DimMember	FactAttendance	1 : many	A member may generate multiple check-ins.
DimMember	FactPayment	1 : many	A member may have multiple payment records.
DimMember	FactSession	1 : many where member linkage exists	Supports member-level booking/session analysis.
DimPlan	FactMembership	1 : many	Filters memberships by plan type.
DimTrainer	FactSession	1 : many	Filters sessions/bookings by trainer.
DimDate	FactAttendance / FactPayment / FactSession / financial facts	1 : many	Provides common period filtering.

Recommended modeling principle: use single-direction relationships from dimensions to facts where practical, and avoid many-to-many relationships unless the source structure genuinely requires them.

4. Model Governance Rules

- Use distinct dimension keys for members, plans, trainers and dates.
- Do not join fact tables directly when a conformed dimension can provide the filter path.
- Validate that joins do not multiply attendance, payment, booking or membership records.
- Use a dedicated date table for period analysis rather than relying only on source-date columns.
- Keep derived flags and business-status logic documented so retention/churn calculations remain explainable.
- Hide technical key columns from end-user report views where they do not add business value.
- Do not publish client-identifiable or sensitive fields in public portfolio versions.

5. Core DAX Measure Specifications

The following measures are portfolio-safe rebuild specifications. Exact source column names must be adapted to the sanitized model.

ID	Measure	Business Meaning	Portfolio DAX Intent
DAX-01	Active Members	Distinct count of members satisfying the agreed active rule for the current filter context.	DISTINCTCOUNT(member) after applying the approved active-member filter.
DAX-02	Eligible Members	Distinct members included in the selected retention/churn denominator.	Distinct count of members meeting the agreed eligibility rules.
DAX-03	Retained Members	Members meeting the agreed retained-member rule for the selected period.	Distinct count of eligible members continuing membership/activity per agreed definition.
DAX-04	Retention Rate %	Percentage of eligible members retained.	DIVIDE([Retained Members], [Eligible Members], 0)
DAX-05	Attrited Members	Members meeting the agreed stopped-attendance and/or cancellation rule.	Distinct count of eligible members satisfying the approved attrition rule.
DAX-06	Attrition Rate %	Percentage of eligible members classified as attrited.	DIVIDE([Attrited Members], [Eligible Members], 0)
DAX-07	Eligible Renewals	Membership records that reached a valid renewal decision point.	Count of memberships eligible for renewal under the selected context.
DAX-08	Renewed Memberships	Eligible memberships that renewed.	Count of eligible membership records with renewal evidence.
DAX-09	Renewal Rate %	Percentage of eligible memberships renewed.	DIVIDE([Renewed Memberships], [Eligible Renewals], 0)
DAX-10	Total Check-ins	Total valid attendance/check-in records.	COUNTROWS(valid attendance records)
DAX-11	Attendance Frequency	Average attendance volume per relevant member.	DIVIDE([Total Check-ins], [Active Members], 0)
DAX-12	Plan Member Count	Distinct members/memberships by plan in current context.	Distinct count using the model grain appropriate to plan analysis.
DAX-13	Total Revenue	Validated revenue in the selected context.	SUM(valid revenue amount)
DAX-14	Total Expenses	Validated expenses in the selected context.	SUM(valid expense amount)
DAX-15	Trainer Sessions	Count of valid trainer-session records.	COUNTROWS(valid session records)
DAX-16	Trainer Bookings	Count of valid class/session bookings.	COUNTROWS(valid booking records) or count of booking IDs

6. Example Portfolio DAX Patterns

These examples illustrate how the measure logic can be implemented after field names are confirmed. They are examples for the reconstructed portfolio model, not verbatim historical client code.

Retention Rate %

```
Retention Rate % =
DIVIDE(
    [Retained Members],
    [Eligible Members],
    0
)
```

Attrition Rate %

```
Attrition Rate % =
DIVIDE(
    [Attrited Members],
    [Eligible Members],
    0
)
```

Renewal Rate %

```
Renewal Rate % =
DIVIDE(
    [Renewed Memberships],
    [Eligible Renewals],
    0
)
```

Total Check-ins

```
Total Check-ins =
COUNTROWS(FactAttendance)
```

Total Revenue

```
Total Revenue =
SUM(FactRevenue[Revenue_Amount])
```

Total Expenses

```
Total Expenses =
SUM(FactExpense[Expense_Amount])
```

7. Cohort Analysis Design

Component	Design
Cohort Key	Joining month derived from the member joining date.
Period Index	Relative month/period since cohort joining period.
Cohort Base	Eligible members in the original joining-month cohort.
Retained / Engaged Count	Members still meeting the selected retention/activity definition in a later cohort period.
Cohort Retention %	Retained/engaged members in cohort period ÷ original eligible cohort size.

Use	Compare joining cohorts longitudinally and identify engagement decline without mixing members of different tenures.
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8. Cohort Measure Control

Because the exact historical inactivity window and retention denominator are not retained, cohort measures should be parameterized or documented with an agreed rule before final portfolio screenshots are generated. The known finding should remain qualitative unless the reconstructed data supports a defensible value.

9. Time & Utilization Measures

Measure	Logic	Business Use
Check-ins by Hour	Count valid check-ins grouped by derived hour/time band.	Identify high-use and lower-use periods.
Check-ins by Day	Count valid check-ins grouped by date/day-of-week.	Understand weekly usage patterns.
Peak Period	Rank or compare time bands by check-in volume.	Highlight comparatively high-demand periods.
Underutilized Period	Identify lower-volume time bands relative to the observed attendance pattern.	Support operational planning.
Member Usage	Check-ins by member over selected period.	Support engagement segmentation.

10. Trainer Measures

Measure	Logic	Control
Trainer Session Count	Count valid session records in trainer context.	Requires reliable Trainer_ID/session linkage.
Trainer Booking Volume	Count valid bookings associated with trainer/session.	Exact booking status logic must be defined from source.
Trainer Attendance Volume	Count realized attendance associated with trainer sessions where linkage exists.	Must avoid double-counting attendance records.
Trainer Workload View	Compare sessions, bookings, assignments and attendance as separate indicators or a documented composite.	Do not invent a weighting if the original formula is unknown.

11. Calculation Validation

- Reconcile distinct member counts before and after model relationships are applied.
- Confirm retention/churn denominators use the intended eligible population.
- Validate renewal eligibility separately from actual renewals.
- Check that time filters affect attendance, revenue, expense and session facts as expected.
- Compare Power BI totals with prepared Python/Excel analysis for selected samples.
- Check that plan filters do not unexpectedly duplicate members with multiple membership records.
- Validate trainer measures against source sessions/bookings before publishing.
- Treat blanks and zero values according to business meaning rather than automatically converting one to the other.

12. KPI Traceability

KPI Catalogue	DAX / Model Support
MEM-01 Member Retention Rate	DAX-02, DAX-03, DAX-04

MEM-02 Attrition / Churn Rate	DAX-02, DAX-05, DAX-06
MEM-03 Renewal Rate	DAX-07 , DAX-08, DAX-09
MEM-04 Active Member Count	DAX-01
MEM-05 Cohort Retention	Cohort key, cohort base and later-period retained/engaged count
UTL-01 to UTL-04	DAX-10, DAX-11 plus hour/day/time-band context
COM-01 to COM-04	DAX-12 to DAX-14
TRN-01 to TRN-03	DAX-15, DAX-16 plus trainer workload indicators

13. Recommended Power BI Model Practices

- Use measures rather than repeated calculated columns for aggregations where appropriate.
- Keep business measures in a dedicated measure table or clearly grouped display folders.
- Use explicit measures for retention, attrition, renewal, attendance and financial KPIs.
- Format rates as percentages and financial measures as currency only after client-safe display requirements are defined.
- Create human-readable measure descriptions to make the model interview- and handover-friendly.
- Use a sanitized portfolio dataset for GitHub screenshots rather than client data.

14. What Is Confirmed vs Reconstructed

Area	Confirmed from Engagement	Portfolio Reconstruction
Power BI	Dashboard built from scratch; filtering and visualization used.	Logical table names and star-schema organization.
Python	Cleaning, EDA, cohort analysis, segmentation, calculations and visualization.	No fabricated historical notebook code.
SQL	SQL used as part of the project toolset.	Exact SQL queries/tables not claimed until recalled or verified.
DAX	Power BI analysis necessarily used model calculations/measures.	Exact original DAX is not retained; measures here are rebuild specifications.
Data Model	Member, membership, attendance, payment, trainer, booking/session, revenue and expense data were analyzed.	Exact historical table names and relationship configuration are not retained.

15. Open Implementation Decisions

- Exact inactivity period that defines stopped-attendance attrition.
- Exact eligibility denominator used for retention and attrition.
- Treatment of members with multiple overlapping membership records.
- Treatment of missing cancellation and renewal records.
- Exact revenue/payment classification and whether payment is a proxy for revenue in any source file.
- Exact trainer-workload calculation if a single index is desired.
- Exact source column names and SQL transformation logic.
- Final sanitized portfolio dashboard page names and layout.